



Staging Phase Transitions in Li_xCoO_2

Zhaohui Chen,^a Zhonghua Lu,^{a,*} and J. R. Dahn^{a,b,*}

^aDepartment of Physics, ^bDepartment of Chemistry, Dalhousie University, Halifax, Nova Scotia B3H 3J5, Canada

Recently, there has been more and more interest in charging LiCoO_2 to potentials above 4.2 V to increase its specific capacity for commercial use. Unlike the situation for Li_xCoO_2 below 4.2 V, the electrochemistry and structure of Li_xCoO_2 above 4.2 V is not completely clear. In this study, two reversible plateaus at 4.55 and 4.63 V were observed in the voltage profile of $\text{Li}/\text{Li}_x\text{CoO}_2$ cells, which were then studied by *in situ* X-ray diffraction (XRD). The intermediate phase, formed at 4.6 V with $x = 0.12$ in Li_xCoO_2 , was determined to be a stage two phase, that is with lithium occupying only every second space between CoO_2 "sandwiches." The structure of this phase was determined using Rietveld refinement of an *ex situ* XRD pattern of $\text{Li}_{0.12}\text{CoO}_2$. We find that adjacent CoO_2 sandwiches are stacked with aa transition metal alignments if there are no Li atoms between them (as in O1- CoO_2) and with ab alignments if there are Li atoms between them (as in O3- Li_xCoO_2). This leads to a six-layer unit cell. The structure we find agrees well with that predicted by Ceder's group for the stage two phase that they call H1-3.

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LiCoO_2 has been the dominant cathode material for commercial Li-ion batteries since 1991. In a commercial cell, LiCoO_2 is usually cycled with an upper cutoff voltage of about 4.2 V with respect to lithium metal. Charge-discharge cycling with this upper cutoff corresponds to repeatedly extracting and inserting about 0.5 Li per LiCoO_2 and gives a specific capacity of about 140 mAh/g. The electrochemistry of $\text{Li}/\text{Li}_x\text{CoO}_2$ cells below 4.2 V and the structural changes that occur in Li_xCoO_2 below 4.2 V are well understood.^{1,2}

To obtain a higher capacity from LiCoO_2 , one must charge it to a potential above 4.2 V to use more than 0.5 Li per LiCoO_2 . However, charge-discharge cycling using a higher upper cutoff voltage often results in rapid capacity loss, thought to be caused by the structural instability of Li_xCoO_2 with $x < 0.5$ and by side reactions with electrolyte at high potentials.

To overcome this rapid capacity loss, Cho *et al.* coated LiCoO_2 with oxides by sol-gel methods.³ Using a ZrO_2 coating, Cho *et al.* showed that LiCoO_2 can deliver a capacity near 170 mAh/g with little capacity loss during the first 70 cycles between 2.75 and 4.4 V.³ Wang *et al.*⁴ and Chen and Dahn⁵ also reported similar effects of coatings on the cycling behavior of Li/LiCoO_2 cells when cycled with upper cutoff potentials of 4.7 and 4.5 V, respectively. Based on these results, LiCoO_2 may be cycled to cutoff potentials of 4.4, 4.5, or even 4.7 V and deliver a specific capacity significantly more than 140 mAh/g.

To use more than 0.5 Li per LiCoO_2 in Li-ion cells, it is important to understand the electrochemical behavior and structural changes of Li_xCoO_2 above 4.2 V. Several groups have done excellent work on this.^{2,6-9} First, Ohzuku and Ueda did *ex situ* X-ray diffraction (XRD) studies on LiCoO_2 electrodes which were charged to different potentials up to 4.8 V.² They observed a phase that formed from the O3 phase when more than 0.75 Li was extracted from LiCoO_2 . However, even when charged to 4.8 V, the O3 phase did not totally transform to the new phase. In the XRD pattern of the purest new phase they obtained there are still some peaks from the O3 phase, such as the (003) and (104) peaks. However, they indexed all the peaks in this pattern with indices of a monoclinic phase except for the (003) peak from the O3 structure and a peak from polyethylene.²

Using dry plastic lithium-ion battery technology, Amatucci *et al.* performed an *in situ* XRD study on a LiCoO_2 electrode which was charged to 5.2 V and then discharged to 3.0 V.⁶ They also observed the formation of the new phase when more than 0.75 Li was re-

moved (initially reported by Ohzuku and Ueda) and they followed Ohzuku's indexing of the powder pattern. In addition, they observed another phase transition at higher potentials. For the first time, they observed that O3- Li_xCoO_2 converted to CoO_2 , with an O1 structure, after charging to 5.2 V.

After the above experimental results were reported, Van der Ven and Ceder calculated the phase diagram of Li_xCoO_2 from first principles.^{7,8} For $0.0 < x < 0.5$ in Li_xCoO_2 , they predicted two phases other than the O3 phase. The phase containing no lithium, at $x = 0.0$, is the O1 phase. This result agrees with the experiments of Amatucci *et al.* Van der Ven and Ceder also predicted that a stage-two compound, named H1-3 by them, would be the most stable phase in the range $0.12 < x < 0.19$. Their calculated XRD pattern of the stage-two phase agrees qualitatively with experimental XRD patterns for the same phase reported by Ohzuku's group and by Amatucci *et al.*⁷

More recently, Sun *et al.* did an *in situ* XRD study on Li_xCoO_2 . From XRD patterns, they observed two phase transitions in the region between $0.0 < x < 0.28$. They agreed that the final phase with $x = 0$ is the O1 phase. However, they did not think that the phase that appears near $x = 0.14$ is a monoclinic phase. They did not work out the structure of this phase and only called it O1a.⁹ It is not possible to describe the diffraction pattern of this phase accurately using the Miller indices that they assigned to the diffraction peaks.

To summarize the above experimental and theoretical work, removing Li from $\text{Li}_{0.5}\text{CoO}_2$ causes the structure of Li_xCoO_2 to change from the O3 phase to a new phase, and then from this phase to the O1 phase. To better understand the cycling of Li_xCoO_2 in the region of $0.0 < x < 0.5$, some questions still must be answered. First, the two-phase transitions were observed structurally during the charge with X-ray diffraction. However, only a single plateau was observed in this region in the voltage-composition curves presented by Amatucci *et al.* Is it possible to observe both phase transitions electrochemically? Second, to use more than 0.5 Li per LiCoO_2 in a commercial cell, one hopes that these structural changes of Li_xCoO_2 are reversible. Are these two phase transitions reversible? Third, there are different opinions about the structure of the phase with $x \approx 0.14$. No profile refinement has been done on XRD pattern of this phase yet. For convenience, we call this phase X-phase temporarily. What is the structure of X-phase?

As shown in our previous study,¹⁰ LiCoO_2 coated with Al_2O_3 can be cycled repeatedly to 4.5 V with little capacity loss in the first 100 cycles. Therefore, Al_2O_3 -coated LiCoO_2 may enable more careful studies of the structural and electrochemical changes of Li_xCoO_2 in the region $0.0 < x < 0.5$. To answer the three questions posed above, an *in situ* XRD experiment was carried out on an

* Electrochemical Society Active Member.

² E-mail: jeff.dahn@dal.ca

Al_2O_3 -coated LiCoO_2 electrode during charge and discharge in the region between $0.0 < x < 0.25$ in Li_xCoO_2 . An *ex situ* XRD pattern of the X-phase was also obtained and a Rietveld refinement was done on this pattern to determine the structure of X-phase.

Experimental

Coating LiCoO_2 with Al_2O_3 .— LiCoO_2 provided by FMC Corp. was used as the powder on which Al_2O_3 coatings were applied. First, 2 g LiCoO_2 was mixed with 20 mL aluminum oxide, polymeric precursor (Alfa Aesar) by magnetic stirring for 3 h. During the mixing, 0.05 g 3-glycidypropyltrimethoxysilane was added as a surfactant. Then the LiCoO_2 powder was separated from the coating solution by centrifugation and was then heated at 100°C for 12 h to evaporate the solvents in the coating solutions. Finally, the heated mixture was reheated at 550°C for 3 h. At present, the amount of coated material and its chemical and physical form in the resultant material is not clear. We do not observe Al_2O_3 by XRD, we do not observe an obvious coated layer by transmission electron microscopy (TEM), and we do not see a shift in lattice parameters as is expected if $\text{Li}[\text{Al}_y\text{Co}_{1-y}]\text{O}_2$ was formed. Based on mass balances during synthesis, we estimate the coated material to be less than 5% by weight compared to LiCoO_2 .

Electrode preparation.—To make electrodes, 500 mg active material was first mixed with 50 mg Super S carbon black (MMM Carbon, Belgium), 125 mg Kynar 2801 (PVDF-HFP), (Elf-Atochem). Then 1.5 g acetone and 200 mg dibutyl phthalate (DBP, Aldrich) were added to the mixture to dissolve the polymer. After extensive mixing, the mixed slurry was spread uniformly on a glass plate using a notch bar spreader. After evaporation of the solvent, the dry film was peeled off the plate and punched into circular disks with a diameter of about 13 mm. The punched electrode was washed several times in anhydrous diethyl ether to remove DBP. The washed electrodes were dried at 100°C in air. A typical electrode disk incorporates about 10 mg of active material.

Electrochemical testing.—To compare the electrochemical behavior of the coated and uncoated LiCoO_2 , electrodes made from both materials were tested in 2325 (23 mm diam, 2.5 mm thickness) coin cells. A polypropylene microporous separator (Celgard 2500), 1 M LiPF_6 in ethyl carbonate (EC): diethyl carbonate (DEC) (33:67 vol) electrolyte (Mitsubishi Chemical), and lithium metal foil were also used within the cell. All cells were assembled in an argon-filled glove box and tested at a constant temperature of $30.0 \pm 0.1^\circ\text{C}$. All cells were cycled using a lower voltage cutoff of 3.6 V and a specific current of 14 mA/g unless specified.

***In situ* XRD.**—The cell for the *in situ* X-ray diffraction measurement is similar to the standard coin cell except that the bottom of the cell was replaced by a Be window to allow the X-ray beam to penetrate to the cathode. To study the structural changes of Li_xCoO_2 during cycling in the region $0.0 < x < 0.25$, the *in situ* cell was first charged to 4.5 V using a current of 10 mA/g. Then it was charged to 4.75 V and discharged to 4.3 V using a current of 1.63 mA/g. The X-ray measurements were made with a Siemens D500 diffractometer equipped with a Cu-target tube and a diffracted beam monochromator. X-ray patterns between 17 and 71° were collected sequentially with a step size of 0.05° in 2θ and each scan took 4 h.

***Ex situ* XRD.**—A cell made with the coated LiCoO_2 as the cathode and Li metal as anode was charged to 4.6 V with an initial current of 14 mA/g. Then the current was stopped for 30 min. After the rest, a current of 75% of the previous current was used to charge the cell to 4.6 V and then it stopped for another 30 min. This cycling regime was repeated twelve times. After this charging, the cell was opened in a helium-filled glove box. The cathode was removed from the cell and sealed in a helium-filled sample holder equipped with an aluminized Mylar window.¹¹ Then, several scans of the X-ray dif-

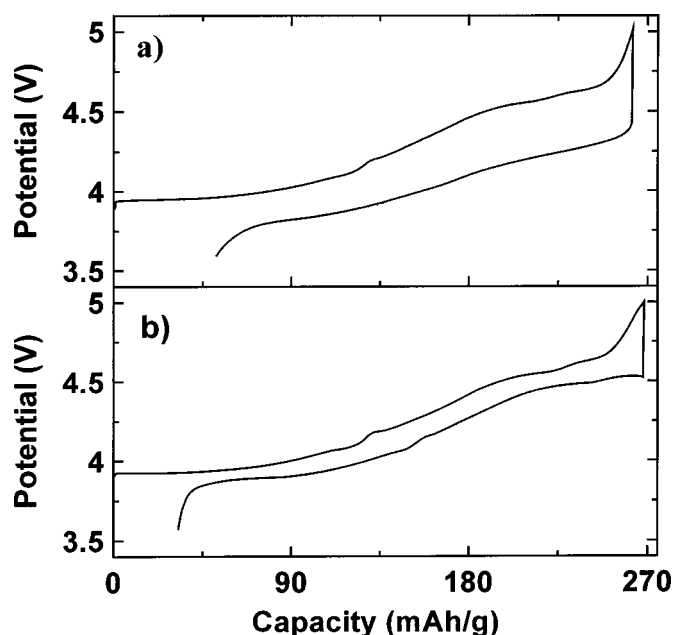


Figure 1. The voltage vs. specific capacity for the first cycle of Li/LiCoO_2 cells with electrodes of (a) the uncoated LiCoO_2 and (b) the coated LiCoO_2 , with an upper cutoff voltage of 5 V.

fraction pattern were made with the Siemens D500 diffractometer. The scans ranged from 8 to 71° in 2θ . A step size of 0.05° in 2θ was used. Each scan took 1 h.

Results and Discussion

Electrochemical behavior of coated and uncoated LiCoO_2 samples when cycled to high potential.—Figure 1 shows the voltage vs. capacity for the first cycle of Li/LiCoO_2 cells using both uncoated and coated LiCoO_2 with an upper cutoff potential of 5 V. The charge parts of both voltage curves are similar and the charge capacities for both materials are close to 270 mAh/g. This indicates

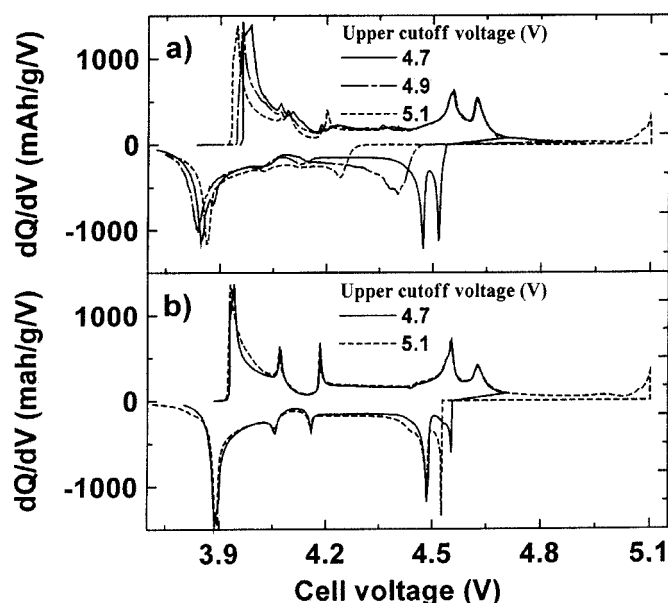


Figure 2. The differential capacity vs. voltage for the first cycle of Li/LiCoO_2 cells with electrodes of (a) the uncoated LiCoO_2 and (b) the coated LiCoO_2 , with different upper cutoff voltages.

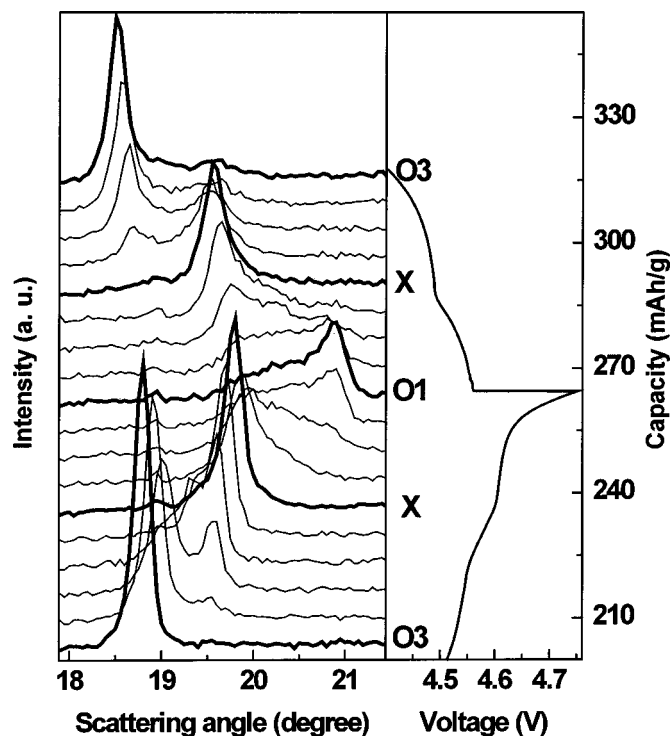


Figure 3. A selected range of the *in situ* XRD patterns as well as the voltage-time curve of the *in situ* cell.

that both materials were almost fully delithiated when charged to 5 V. In both charge curves, one can see two small plateaus corresponding to the order-disorder phase transition near 135 mAh/g and the two plateaus at 4.55 and 4.63 V which correspond to the two phase transitions from the O3 phase to the X-phase and then to the O1 phase.

However, the discharge portions of the voltage-capacity curves for the two samples are quite different. For the uncoated LiCoO_2 , it is difficult to observe any plateaus in the discharge voltage profile.

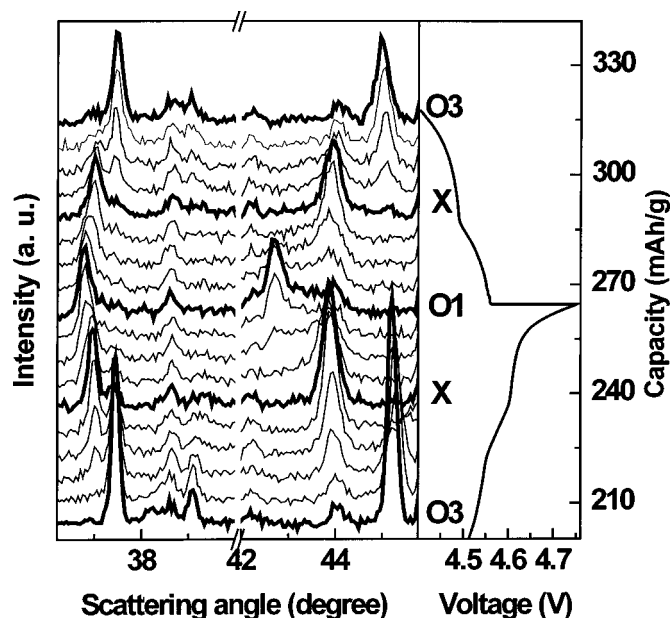


Figure 4. A selected range of the *in situ* XRD patterns as well as the voltage-time curve of the *in situ* cell.

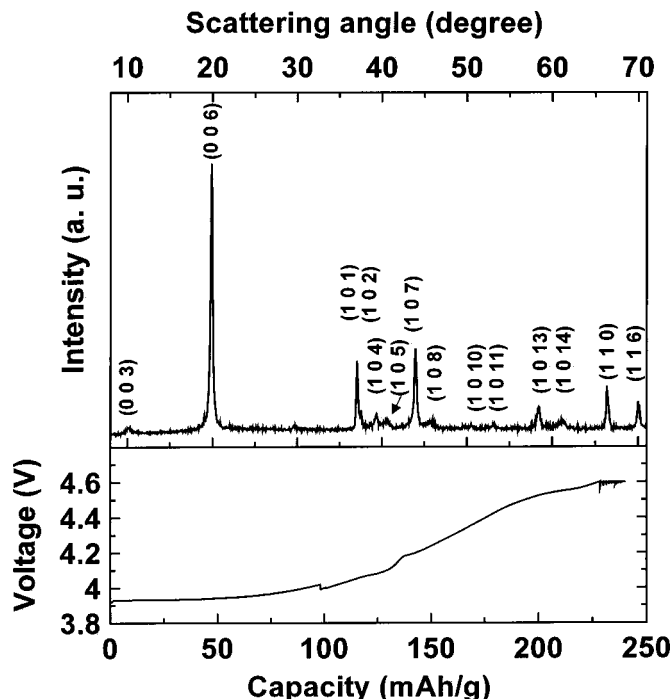


Figure 5. (top) The *ex situ* XRD pattern indexed with the six-layer structure. (bottom) The voltage-capacity curve of the *ex situ* cell used to prepare the sample for collecting the pattern above.

The cell voltage dropped from 5 V to about 4.3 V immediately after the discharge began. The average potential of the discharge curve is much lower than the average potential of the charge curve. This experiment shows that uncoated LiCoO_2 cannot be cycled reversibly with an upper cutoff voltage of 5 V. On the contrary, the discharge curve of the coated LiCoO_2 shows much better reversibility. All the plateaus observed during charge have counterparts in the discharge curve of the coated sample.

When the differential capacity, dQ/dV , is plotted vs. the cell potential, plateaus in voltage-capacity curves appear as peaks in dQ/dV . Figure 2 shows dQ/dV vs. voltage for the first cycle of Li/LiCoO_2 cells having positive electrodes of uncoated LiCoO_2 and coated LiCoO_2 . Different upper cutoff voltages were employed. To

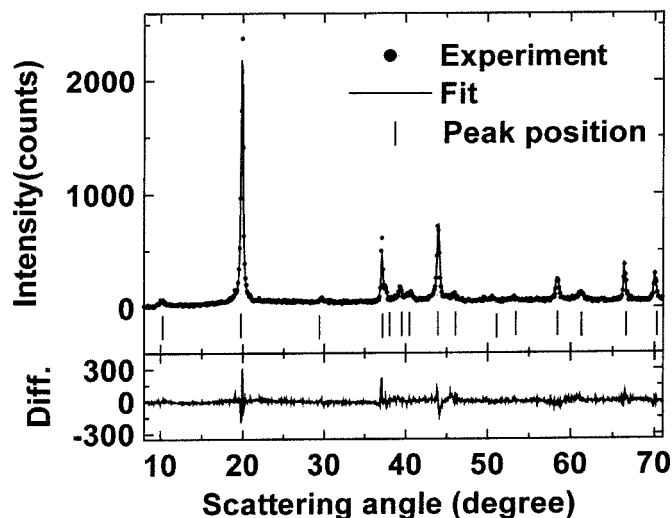


Figure 6. A comparison of the *ex situ* XRD pattern and the calculated pattern from the best Rietveld refinement.

Table I. Experimental and calculated peak positions for the diffraction pattern of the H1-3 phase.^a

<i>h</i>	<i>k</i>	<i>l</i>	2θ observed (degree)	2θ calculated (degree)	Difference (degree)
0	0	3 ^b	9.9	9.8	
0	0	6	19.777	19.719	0.058
1	0	1	36.933	36.960	-0.028
1	0	2	37.432	37.424	0.008
1	0	4	39.220	39.230	-0.010
0	1	5 ^b	40.4	40.5	
1	0	7	43.817	43.874	-0.057
0	1	11 ^b	52.9	52.8	
1	0	13	58.287	58.305	-0.017
1	0	14 ^b	61.0	61.1	
1	1	0	66.241	66.235	0.006
1	1	6	69.893	69.852	0.041

^a The lattice constants used in the calculation were $a = 2.823 \text{ \AA}$ and $c = 27.07 \text{ \AA}$.

^b Peak not used for the least-square fitting due to its weak intensity and uncertain peak position.

make the features in high voltage parts easy to see, the intense peaks near 3.9 V have been cut off for both materials. Figures 2a and b show that all the charge curves have two peaks at 4.55 and 4.63 V, respectively. These two peaks, corresponding to the highest two plateaus in the charge curves in Fig. 1, represent the phase transitions from the O3 phase to the X-phase and then to the O1 phase. Therefore, these two phase transitions can be observed electrochemically.

Figure 2 suggests that if Li/LiCoO₂ cells are charged to 4.5 V, then Li_xCoO₂ is still in the O3 phase. If the cells are charged to 4.6 V, Li_xCoO₂ is in the X-phase. If the cells are charged to 4.7 V, Li_xCoO₂ is mostly in the O1 phase. Further increases to the upper cutoff potential did not uncover further reversible capacity. When the cells are charged to above 5 V, the dQ/dV curve shows another peak for both materials. This peak is caused by electrolyte decomposition at high potential.

There are two peaks in the discharge part of some dQ/dV curves that correspond to the reverse reactions of the 4.55 and 4.63 V peaks in the charge curves. This indicates that the two phase transitions are reversible under certain conditions. When the uncoated LiCoO₂ is charged to above 4.8 V, the two phase transitions are no longer reversible. The two sharp peaks near 4.5 V in the discharge curve become one broad peak. As the upper cutoff voltage increases to 5.1 V, the broad peak shifts to lower potential and becomes weaker. By contrast, for the Li/LiCoO₂ cell with the coated LiCoO₂ electrode, one can still observe the two peaks in its discharge dQ/dV vs. voltage curve even after charging to 5.1 V. This indicates that the surface coating with Al₂O₃ improves the stability and reversibility of LiCoO₂ during cycling to high cutoff potentials. Therefore, we decided to use coated LiCoO₂ for *in situ* XRD studies.

***In situ* XRD of coated LiCoO₂.**—As shown above, the two phase transitions from the O3 phase to the X-phase and then to the O1 phase during a charge are reversible for LiCoO₂ coated with Al₂O₃. Using the *in situ* X-ray diffraction technique, one can see this reversible process more directly. Figures 3 and 4 show the XRD patterns of Li_xCoO₂ coated with Al₂O₃ taken between 4.5 and 4.75 V during charge and between 4.75 and 4.4 V during discharge. In Fig. 3, during charge, the (003) peak from the O3 phase gradually disappears in the beginning, and a peak from the X-phase simultaneously grows. After this phase transition is finished, the peak from the X-phase diminishes, while the (001) peak from the O1 phase grows. At 4.75 V, one can see the (001) peak from the O1 phase in the corresponding XRD pattern even though there is a broad shoulder on its left side that indicates the phase transition is not complete

yet. During discharge, the phase transitions occur reversibly. Figure 4 shows another part of the *in situ* XRD patterns. It also clearly shows these reversible phase transitions.

Amatucci *et al.* also did *in situ* studies during a discharge of Li/Li_xCoO₂ cells. In their case, the cell voltage dropped to 4.35 V as soon as the *in situ* cell was placed on discharge after charging to 5.2 V.⁶ That phenomenon is similar to the one observed in Fig. 2a for uncoated materials. In Fig. 2a, for the cell made with uncoated LiCoO₂ and charged to 5.1 V the cell voltage dropped to 4.32 V as soon as the discharge began. The two phase transitions from the O1 phase to the X-phase and then to the O3 phase are not apparent in the dQ/dV vs. voltage data. Therefore, it is not surprising that Amatucci *et al.* did not observe these two phase transitions during the discharge of uncoated LiCoO₂ by *in situ* XRD.

Determination of the structure of X-phase by *ex situ* XRD and Rietveld refinement.—Compared to the XRD patterns for X-phase from the *in situ* results of Amatucci *et al.* and Sun *et al.*, the corresponding pattern in this *in situ* study is similar. It also qualitatively agrees with the calculated pattern of the H1-3 phase from Ceder's work. This indicates that X-phase possibly has an H1-3 structure. To further confirm this, an *ex situ* XRD pattern was obtained from an electrode charged to 4.6 V as described in the Experimental section.

Figure 5 at the bottom shows the voltage profile of the cell that was used for the *ex situ* analysis. The voltage drop at a capacity of about 100 mAh/g was caused by an interruption of the charge current. The plateau between 4.5 and 4.6 V indicates that the O3 phase had transformed to the X-phase. Our best estimate of the composition of the X-phase is Li_{0.12}CoO₂ based on the results in Fig. 3-5. Figure 5 at the top shows the XRD pattern obtained by adding the first three *ex situ* scans together to obtain a better signal-to-noise ratio. According to Van der Ven *et al.*, the structure of X-phase should be a stage-two structure. In the stage-two structure, lithium atoms in every other Li layer in the O3 structure of Li_xCoO₂ are totally removed. According to Van der Ven, the adjacent CoO₂ sandwiches now shift to the stacking found in the O1 phase while the CoO₂ sandwiches that are still separated by lithium retain the original stacking arrangement found in the O3 phase. Since this is a "hybrid" of the O1 and O3 structures, Van der Ven *et al.* called it the H1-3 structure.

In the notation of hexagonal close-packed layers the stacking of the oxygen (capital letters), cobalt (lower case letters), and lithium atoms (Greek symbols) in the H1-3 phase is



where the vertical bars (|) span the repeating unit. Thus six CoO₂ slabs are needed to form a unit cell of this structure. Figure 5 at the top shows that this *ex situ* XRD pattern can be well indexed as a single H1-3 phase. The lattice constants of this phase determined by least-square fitting to the measured positions of the Bragg peak positions of the main peaks are $a = 2.823 \pm 0.002 \text{ \AA}$ and $c = 27.07 \pm 0.03 \text{ \AA}$. Table I shows a comparison between the

Table II. Rietveld refinement results for the *ex situ* XRD pattern.^a

$a = 2.8193 \pm 0.0004 \text{ \AA}$		$c = 27.035 \pm 0.011 \text{ \AA}$		
$R_B = 5.7\%$		$R_{WP} = 17.55\%$		
	Site	<i>x</i>	<i>y</i>	<i>z</i>
Li	3a	0	0	0
Co	6c	0	0	0.42150 ± 0.00016
O1	6c	0	0	0.27671 ± 0.00045
O2	6c	0	0	0.11535 ± 0.00045

^a The space group used was $R\bar{3}m$. The occupancy for Li is 0.24 and the occupancy for Co, O1, and O2 is 1.

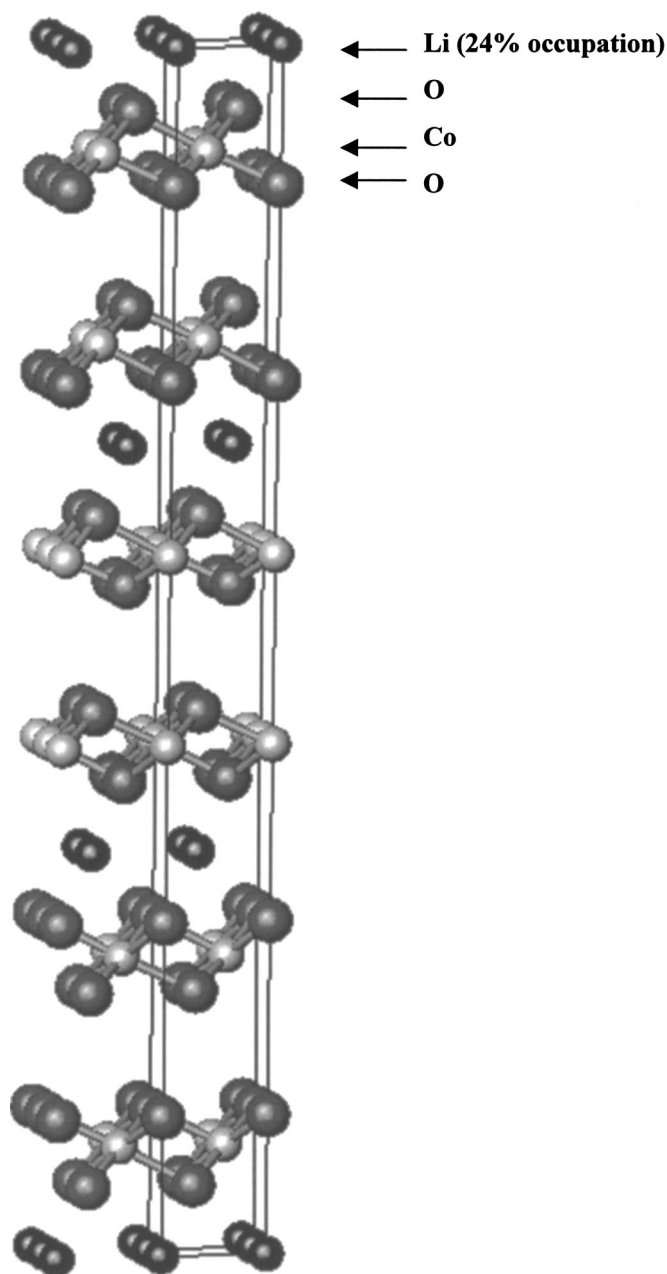


Figure 7. A view of the H1-3 structure calculated using the results of the best Rietveld refinement (only about 24% of the Li sites in the Li-containing planes are occupied).

measured peak positions and the calculated ones assuming the H1-3 structure.

The H1-3 phase takes the $R\bar{3}m$ space group with Li occupying 3a sites, and Co and the two oxygen atoms in three different 6c sites. With this structural information and the lattice constants obtained above, a Rietveld refinement was performed for this *ex situ* XRD pattern. The results are listed in Table II. Figure 6 shows the *ex situ* XRD pattern and the calculated pattern based on the best Rietveld fit. It was noticed that the *ex situ* XRD patterns from consecutive 1 h scans were slightly different, which means that the H1-3 phase is not very stable even when sealed in the helium-filled holder. Taking this into account, the refinement of the *ex situ* XRD pattern with the H1-3 structure is very good. Based on this refinement, we can confirm that X-phase has an H1-3 structure as predicted by Van der Ven

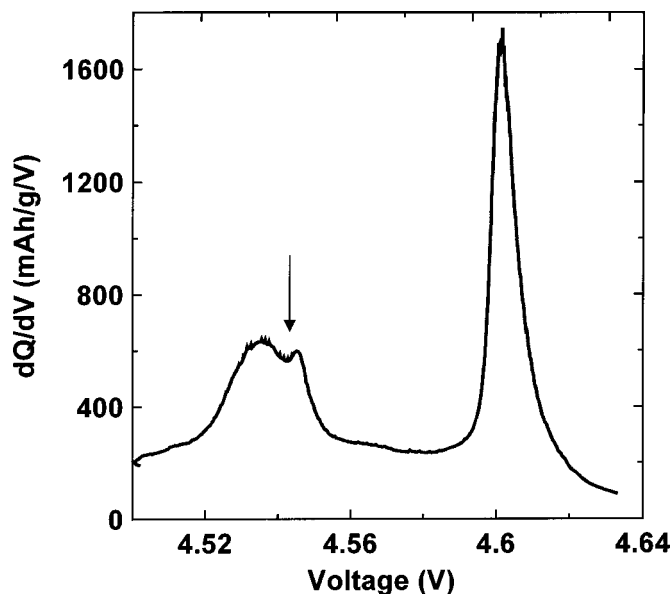


Figure 8. The differential capacity vs. voltage of Li/LiCoO₂ cell made with the coated LiCoO₂ and charged with a current of 0.5 mA/g.

et al. Based on the results of the refinement, Fig. 7 was made to show the atom positions in the H1-3 structure.

More staged phases?—It was confirmed above that there is a staging phase transition in Li_xCoO₂ during the charge of Li/LiCoO₂ cells to near 4.6 V. In addition to stage-two, does any other staged phase appear? A careful investigation of Fig. 3 reveals some interesting features. In the region of the phase transition from the O3 phase to the H1-3 phase during the charge, the two XRD patterns before the H1-3 phase was totally formed show a small peak in between the (003) peak from O3 and the (006) peak from H1-3. This indicates that there may be another staged phase formed.

To observe further evidence for this multiple staging, a Li/LiCoO₂ cell was charged with a current of 0.5 mA/g above 4.5 V. Figure 8 shows the dQ/dV vs. voltage curve for this cell. It shows the splitting of the peak representing the phase transition from the O3 phase to the H1-3 phase observed before. Therefore, there may be multiple phase transitions during the transition from the O3 phase to the H1-3 phase during the charge. However, to be sure about this multiple staging phenomenon, even more careful work is needed.

Conclusions

For the first time, the two plateaus in the voltage-capacity curves of Li/LiCoO₂ cells above 4.5 V, representing the sequential phase transitions from the O3 phase to the H1-3 phase and from the H1-3 phase to the O1 phase, have been reported. With an upper cutoff voltage of 4.7 V, but not 4.8 V or above, the two phase transitions are reversible. However, if the LiCoO₂ is coated with Al₂O₃, then the two phase transitions are reversible even with an upper cutoff voltage of 5.1 V. *In situ* XRD experiments also show these two phase transitions during a charge of the coated LiCoO₂ to 4.75 V and their corresponding ones during the subsequent discharge. A Rietveld refinement of an *ex situ* XRD pattern obtained from an electrode charged to 4.6 V confirmed that this is the stage-two compound proposed by Van der Ven *et al.* from first principles. We believe the composition of this phase is Li_{0.12}CoO₂. In addition, *in situ* XRD results and differential capacity measurements of a Li/LiCoO₂ cell charged with a specific current of only 0.5 mA/g suggest evidence for the possibility of a multiple staging phenomenon during the charge of LiCoO₂.

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